WORKED SOLUTION

STEP II 1989, Question 10

Pure • Lines and Planes

STEP II 1989, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q10

The key idea

Two lines in space can fail to meet in a single point in three ways: they can be the same line, they can be parallel, or they can be *skew*, missing each other in different planes. So "exactly one point" needs two conditions: the directions are not parallel, and the lines lie in a common plane. The cross product handles both at once. For the planes, the line of intersection runs along the cross product of the normals, and the final part is the first part again in disguise.

Solution

First part: when two lines meet in exactly one point

The lines meet where $\mathbf{a} + \lambda\mathbf{u} = \mathbf{b} + \mu\mathbf{v}$, that is, where

$$\lambda\mathbf{u} - \mu\mathbf{v} = \mathbf{b} - \mathbf{a}.$$

For these to be genuine lines we need $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$. Then:

- If $\mathbf{u}$ and $\mathbf{v}$ are parallel ($\mathbf{u} \times \mathbf{v} = \mathbf{0}$), the lines are either parallel and distinct (no common point) or identical (infinitely many). Either way, not exactly one.
- If $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$, then $\mathbf{u}$ and $\mathbf{v}$ span a plane, and the equation has a solution exactly when $\mathbf{b} - \mathbf{a}$ lies in that plane, i.e. is perpendicular to $\mathbf{u} \times \mathbf{v}$. When a solution exists it is unique, because $\mathbf{u}$ and $\mathbf{v}$ are independent.

So the conditions are

$$\mathbf{u} \neq \mathbf{0}, \quad \mathbf{v} \neq \mathbf{0}, \quad \mathbf{u} \times \mathbf{v} \neq \mathbf{0}, \quad (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{u} \times \mathbf{v}) = 0.$$

(The first two are implied by the third, but it's worth saying them.) If the last condition fails, the lines are skew.

Second part: the two planes

The normals are $\mathbf{n}_1 = (1, 1, b)$ and $\mathbf{n}_2 = (b, b, 1)$, and

$$\mathbf{n}_1 \times \mathbf{n}_2 = \left(1 - b^2,\ b^2 - 1,\ 0\right) = (1 - b^2)(1, -1, 0).$$

Two planes fail to meet in a line exactly when their normals are parallel, which happens here when $1 - b^2 = 0$, so $b = 1$ or $b = -1$.

- $b = 1$: both equations become $x + y + z = 3$.
- $b = -1$: the equations are $x + y - z = 1$ and $-x - y + z = -1$, the same plane again.

So in both cases the two planes coincide, rather than being parallel and separate.

Now take $b \neq \pm 1$. The line runs along $\mathbf{n}_1 \times \mathbf{n}_2$, so we can use $\mathbf{u} = (1, -1, 0)$. To find points on it, notice that x and y appear only through $s = x + y$:

$$s + bz = b + 2, \qquad bs + z = 2b + 1.$$

Subtracting b times the first from the second gives $(1 - b^2)z = 1 - b^2$, so $z = 1$ (we may divide since $b^2 \neq 1$), and then $s = 2$. The line of intersection is therefore $x + y = 2$, $z = 1$, independent of b.

We need a point $\mathbf{a}$ on this line with $\mathbf{a} \cdot \mathbf{u} = 0$, i.e. $x - y = 0$. With $x + y = 2$ this gives $x = y = 1$, so

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad b \neq \pm 1.$$

The point $\mathbf{a}$ is the foot of the perpendicular from the origin to the line.

Final part

The points whose coordinates satisfy both equations $(*)$ form the line above when $b \neq \pm 1$, and a whole plane when $b = \pm 1$. We treat the cases separately.

? **Ask yourself** What set of points satisfies both equations when $b = \pm 1$?

It is a whole plane, not the empty set, because we found that the planes coincide. That changes the answer completely in those cases.

Case $b \neq \pm 1$. We need the given line to meet the line $\mathbf{r} = (1, 1, 1) + \lambda(1, -1, 0)$ in exactly one point. Apply the first part with $\mathbf{a} = (1, 1, 1)$, $\mathbf{u} = (1, -1, 0)$, $\mathbf{b} = (0, 0, c)$, $\mathbf{v} = (1, d, 0)$. Then

$$\mathbf{u} \times \mathbf{v} = (0,\ 0,\ d + 1), \qquad (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{u} \times \mathbf{v}) = (-1, -1, c - 1) \cdot (0, 0, d + 1) = (c - 1)(d + 1).$$

We need $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$, so $d \neq -1$, and then $(c - 1)(d + 1) = 0$ forces $c = 1$. So the conditions are $c = 1$ and $d \neq -1$. (Directly: the given line has $z = c$, so it meets $z = 1$ only if $c = 1$, and then $x + y = \mu(1 + d) = 2$ has the single solution $\mu = 2/(1 + d)$ when $d \neq -1$.)

Case $b = \pm 1$. Now we need the given line to meet a plane in exactly one point. A point $(\mu, d\mu, c)$ lies on $x + y + z = 3$ (for $b = 1$) when $\mu(1 + d) = 3 - c$, and on $x + y - z = 1$ (for $b = -1$) when $\mu(1 + d) = 1 + c$. Each equation has exactly one solution for μ if and only if $1 + d \neq 0$; if $d = -1$ the line is parallel to the plane and there are no solutions or infinitely many. Here c can be anything.

Answer. There is exactly one such point if and only if

$$d \neq -1 \quad \text{and} \quad \big(c = 1 \text{ or } b = \pm 1\big),$$

which can be written compactly as $d \neq -1$ and $(b^2 - 1)(c - 1) = 0$.

Checks

- Substitute $(1 + \lambda, 1 - \lambda, 1)$ into $(*)$: $2 + b = b + 2$ and $2b + 1 = 2b + 1$, for every b and λ.
- With $b = 2$, $c = 1$, $d = 3$ the point is $\mu = \frac{1}{2}$, i.e. $(\frac{1}{2}, \frac{3}{2}, 1)$: $\frac{1}{2} + \frac{3}{2} + 2 = 4 = b + 2$ and $1 + 3 + 1 = 5 = 2b + 1$.
- A computer tested every b, c, d in $\{-3, \ldots, 3, \frac{1}{2}, -\frac{5}{3}\}$ in exact arithmetic (729 cases), solving for μ directly, and the count was one exactly when the stated conditions hold.

Common slips

- Saying that two lines meet in one point just because they aren't parallel. In three dimensions they can be skew.
- Assuming that when $b = \pm 1$ the planes are parallel and separate, and concluding there are no points at all. Check the right-hand sides: the planes coincide.
- Dividing by $1 - b^2$ without saying why it is non-zero.

Beyond STEP

This is a small system of linear equations in disguise: two equations in three unknowns have a line of solutions when the coefficient matrix has rank 2, and a plane (or nothing) when it has rank 1. Adding the line's parametrisation turns it into an overdetermined system in μ, and "exactly one solution" is a statement about rank and consistency. Strang's *Introduction to Linear Algebra* builds all of this from the row picture (planes meeting) and the column picture (combinations of vectors) that appear here.

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